

Path-dependent Approach to Volatility: Analysis of VIX Futures Dynamics

Author: Ilia Demskii

Supervisor: Yaroslav Lyulko **University:** HSE, ICEF **Academic Year:** 2025–2026

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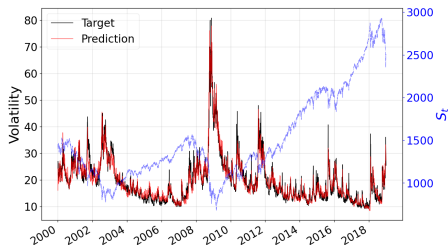
- ① Guyon's Path-Dependent Volatility (PDV) model
- ② From Fitting to Forecasting: Motivation and Research
- ③ Two Forecasting Approaches: SDE and CatBoost
- ④ Forecasting and PnL simulation results: SDE and CatBoost
- ⑤ Additional Contributions: Theory add-on
- ⑥ Conclusion and Q&A

The Guyon–Lekeufack PDV model

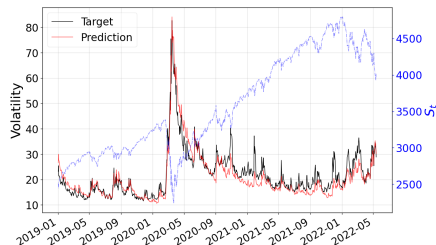
For a given volatility index σ and market index S , the model estimates volatility as

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, \quad \beta_0 > 0, \beta_1 < 0, \beta_2 \in (0, 1),$$

$$R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad R_{2,t} = \sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2, \quad r_{t_i} = \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}}.$$

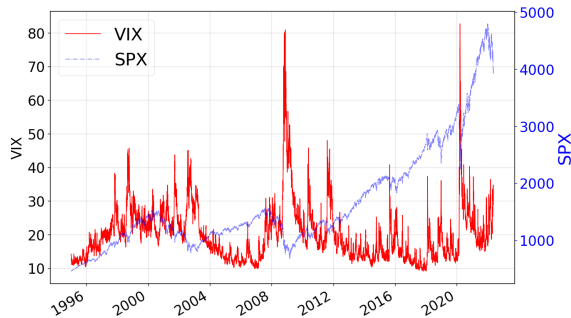


Train (2000–2018) $R^2_{\text{train}} = 0.947$



Test (2019–2022) $R^2_{\text{test}} = 0.862$

Stylized facts of volatility (SPX & VIX)



- **Leverage effect:**
 $\text{Corr}(r_t, \sigma_{t+h}) < 0, h > 0.$
- **Volatility clustering:**
high- and low-vol regimes cluster in time.
- **Volatility roughness:**
 $\mathbb{E}|\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \sim c \Delta^{qH}, H \approx 0.15.$
- **Weak Zumbach effect:**
$$\text{Corr}\left(\left(\frac{dS_t}{S_t}\right)^2; \sigma_{t+h}\right) > \text{Corr}\left(\sigma_t; \left(\frac{dS_{t+h}}{S_{t+h}}\right)^2\right)$$
- **Strong Zumbach effect:**
Law($\sigma_{t+h} \mid \mathcal{F}_t$) depends on past returns, not only past σ .

Markovisation: TSPL $\rightarrow$ 2-EXP $\rightarrow$ SDE

(1) Continuous-time PDV setup

$$\frac{dS_t}{S_t} = \sigma_t dW_t, \quad \sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}.$$

(2) TSPL memory kernels

$$K_{\alpha,\delta}(\tau) = Z_{\alpha,\delta}^{-1}(\tau + \delta)^{-\alpha}, \quad Z_{\alpha,\delta} = \frac{\delta^{1-\alpha}}{\alpha - 1}.$$

$$R_{1,t} = \int_{-\infty}^t K_1(t-u) \frac{dS_u}{S_u}, \quad R_{2,t} = \int_{-\infty}^t K_2(t-u) \left(\frac{dS_u}{S_u} \right)^2.$$

(3) Markovianization (2-EXP approximation)

$$K_n(\tau) = (1 - \theta_n) \lambda_{n,0} e^{-\lambda_{n,0} \tau} + \theta_n \lambda_{n,1} e^{-\lambda_{n,1} \tau}.$$

(4) 4-factor Markovian PDV

$$dR_{1,j,t} = \lambda_{1,j} (\sigma_t dW_t - R_{1,j,t} dt),$$

$$dR_{2,j,t} = \lambda_{2,j} (\sigma_t^2 - R_{2,j,t}) dt, \quad j \in \{0, 1\},$$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t.$$

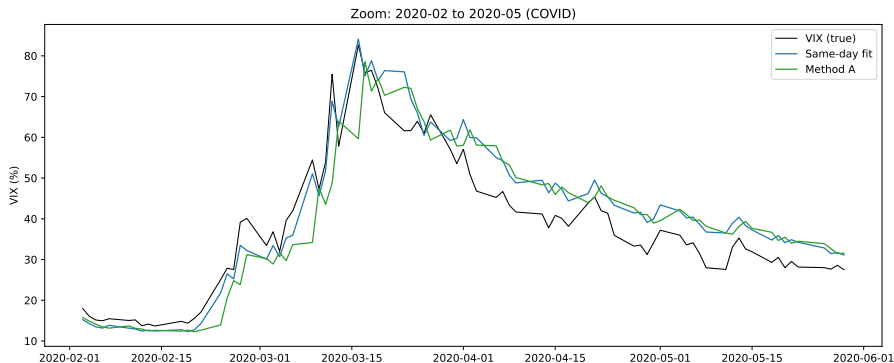
State $(S_t, \sigma_t, R_{1,j,t}, R_{2,j,t})$ is Markov.

These, together with previous results, allowed to solve the open joint SPX–VIX implied-volatility calibration problem and earned Julien Guyon Quant of the Year 2025 award by Risk.net.

Motivation & Research Question

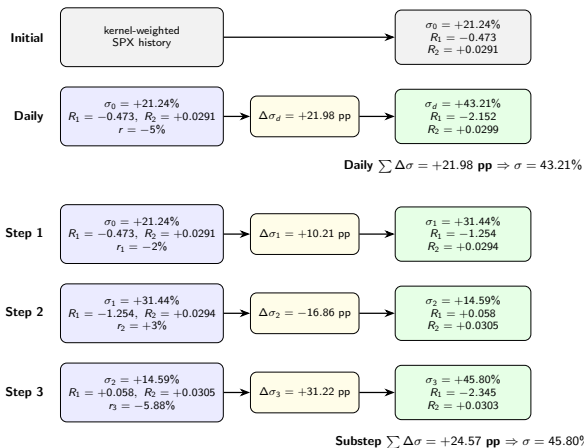
- **Observation:** the PDV model fits VIX:SPX well empirically. But does it predict?
- **Research Question:** Does the path-dependent volatility model provide a statistical edge in short-term VIX-futures returns forecasting?
- **Hypothesis:** past returns and features carry predictive signal for VIX futures at the 1-minute frequency.
- **Task:** Re-frame and modify the PDV model into a short-term forecasting machine. Validate via directional accuracy, MAE and simulated PnL.

Error analysis I: Recent returns



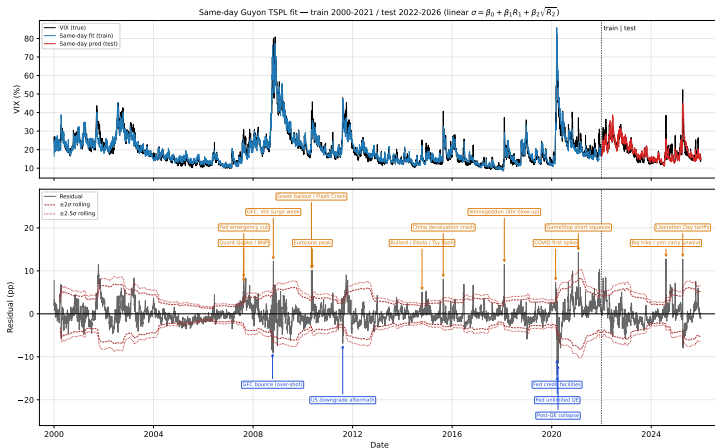
The kernel puts $W_1 = 4.28\%$, $W_2 = 4.82\%$ on the latest return (of 1000 days). Withhold it and the fit lags as a copy of itself. Yields little power on a daily scale.

Error analysis II: Intraday Trajectory Matters



Same daily return, apart by 2.59%. Typical bid-ask spread for VIX futures is 0.05%

Error analysis III: R_2 saturation & other Residuals



Reducible

- 1 Foreign shocks
- 2 R_2 saturation

Irreducible

- 1 Fed / government policy
- 2 News
- 3 Agents' actions

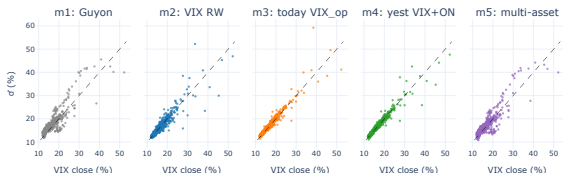
Error analysis IV: Verification

Test 2023-07 – 2025-09

#	Model	Mechanism	RMSE	R^2	DC
1	Original Model	$VIX_t = \beta_0 + \beta_1 R_1^{SPX} + \beta_2 \sqrt{R_2^{SPX}}$	2.61%	0.69	0.55
2	VIX random walk	$VIX_t = VIX_{t-1}$	1.96%	0.82	0.50
3	Today VIX_{open} + intraday	$VIX_t = VIX_t^{open} + \sum \Delta VIX$	1.46%	0.90	0.71
4	Random Walk + overnight + intraday	$VIX_t = VIX_{t-1} + \sum \Delta VIX$	1.32%	0.92	0.73
5	Multi-index Guyon	$VIX_t = \beta_0 + \sum_X (\beta_{1,t}^X R_{1,t}^X + \beta_{2,t}^X \sqrt{R_{2,t}^X})$	2.34%	0.75	0.61

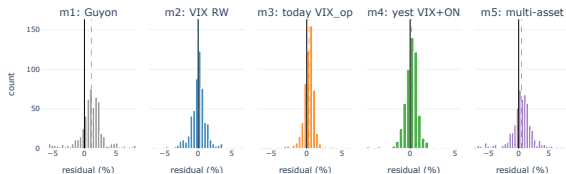
$X \in \{SPX, SX5E, NKY\}$.

(e) Fit vs actual — scatter



(10e) fit vs. actual, per model

(f) Residual distribution



(11f) residual distribution, per model

Forecasting I: SDE approach

Model used: 2EXP PDV

$$\frac{dS_t}{S_t} = r_t$$

$$dR_{1,j,t} = \lambda_{1,j}(r_t - R_{1,j,t})dt,$$

$$dR_{2,j,t} = \lambda_{2,j}(r_t^2 - R_{2,j,t})dt,$$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t$$

Signal generation

$$\mathbb{E}[d\sigma_t \mid \mathcal{F}_t] = \mu_t dt$$

$$f_t(dt) = 100 \mu_t dt \quad dt \in \{1, 5, 10, 15, 30, 60\} \text{ min}$$

Trade 10:30–15:00 ET;

Enter signal if in the top of 5/10/20% of $|f_t(dt)|$

Data and Propagation

Data. CBOE daily SPX/VX OHCLV (07-2008–02-2025);
Databento ES/VX MBP-10 (03-2025–08-2025).

Calibrate. Fit 2EXP params on SPX/VX, 2008–2025, obtain
initialisation point. Train $R^2 = 0.893$.

Propagate. On last day of train initialize 2EXP model and
advance 1-min on ES/VX mid (RTH+ETH).

Validate. SDE requires no parameters to recalibrate. Out-of-
sample is whole 03-2025–08-2025.

Execution

Enter. Post a limit at the best touch; join the back of the
queue.

Monitor. Drain the queue on MBP-10 events through $t+dt$.

Close. +1-tick take-profit if possible, o/w force-exit at $t+dt$.
If unfilled remove order w/o fee; shorts mirror.

Forecasting II: CatBoost approach

Compare to ML approach by incorporation of order-book microstructure.

Features

- **ES block:**
Order-book imbalance, microprice deviation, spread, log-returns, realized vol, MACD.
- **VX block:**
Order-book imbalance, microprice deviation, spread, log-returns, realized vol, MACD.
- **PDV block:**
 $R_1\text{-drift} = -\beta_1 \bar{\lambda}_1 \bar{R}_{1,t},$
 $R_2\text{-drift} = \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}}.$
- **Time block:**
Session progress.

Target

$\Delta VX_t^h = VX_{t+h} - VX_t$ under L_2 loss
Analog of Guyon's $\mathbb{E}[d\sigma_t | \mathcal{F}_t]$
Allows for direct SDE-vs-ML comparison.

Why CatBoost?

- **Model free:**
Ingests all 15 microstructure features directly
- **Scale-free**
Doesn't require normalization of data.
- **Non-linear**
Able to catch non-linearities between data.
- **Symmetric oblivious trees**
Regularized by construction, less overfit.
- **Ordered boosting**
Leakage-resistant gradient estimates.

Validation

Walk-forward: 7 train / 2 val / 1 test / 1 slide day
Folds within ± 3 days of a VX settlement rejected
OOS pooled.

CatBoost: Feature Importance

- 1 At 40–60s the VX book holds $\sim 74\%$ of importance.
- 2 Weight shifts to the R_1 -drift as h grows; the VX book falls below 30% by 600s.
- 3 The two PDV drifts cross over: R_2 -drift leads at short h , R_1 -drift dominates at long h
- 4 ES stays at 5–8% throughout.

Walk-forward importance by block (%)

h (s)	ES	VX	R_1 -drift	R_2 -drift	Time
40	8.2	73.5	3.0	12.4	2.9
60	5.4	73.9	4.7	12.7	3.3
300	8.4	30.5	45.1	10.3	5.7
600	7.1	23.9	53.7	8.7	6.6
900	6.3	15.4	63.3	8.1	6.9

Forecasting performance: SDE vs. CatBoost

SDE approach

$h(s)$	p_{50}	p_{90}	p_{95}	p_{99}	spread
60	0.0001	0.0003	0.0006	0.0027	0.05
300	0.0003	0.0017	0.0033	0.0161	0.05
600	0.0007	0.0037	0.0070	0.0291	0.05
900	0.0010	0.0057	0.0111	0.0576	0.05

forecast magnitude $|f_t(H)|$: deeply subspread

$h(s)$	MAE ₁	MAE ₀	Dir. acc.	DM (p)	$n_{\neq 0}$	n
60	0.032	0.032	0.490	+1.13 (0.258)	10,574	26,573
300	0.073	0.073	0.485	+0.97 (0.332)	3,357	5,394
600	0.105	0.105	0.479	+1.03 (0.305)	1,951	2,746
900	0.132	0.131	0.500	+0.75 (0.454)	1,396	1,864

forecasting skill: no edge

CatBoost approach

$h(s)$	p_{50}	p_{90}	p_{95}	p_{99}	spread
40	0.0005	0.0033	0.0144	0.0553	0.05
60	0.0006	0.0037	0.0053	0.0514	0.05
300	0.0028	0.0274	0.0419	0.2281	0.05
600	0.0047	0.0492	0.0871	0.4402	0.05

forecast magnitude: sub-spread improvement

$h(s)$	MAE ₁	MAE ₀	Dir. acc.	perm.	DM (p)	$n_{\neq 0}$	n
40	0.044	0.043	0.589	0.511	+1.72 (0.085)	4,620	10,721
60	0.053	0.052	0.577	0.510	+2.80 (0.005)	3,684	7,717
300	0.111	0.107	0.549	0.517	+2.52 (0.012)	1,315	1,984
600	0.163	0.155	0.510	0.501	+2.22 (0.027)	777	1,061
900	0.184	0.180	0.521	0.504	+0.97 (0.330)	560	732

forecasting skill: sign edge at sub-minute

PnL Simulations: SDE vs. CatBoost

SDE (top-5% gate)

h	n_{fire}	$P(\text{fill} \mid \text{move})$			ΔPnL		PnL
		adv	fav	ratio	adv	fav	
60s	1,329	0.68	0.68	1.00	-78	-86	-70,301
300s	270	0.90	0.70	1.29	-165	+12	-16,958
600s	138	0.91	0.78	1.16	-206	+42	-10,860
900s	94	1.00	0.80	1.24	-228	+45	-7,135
30m	50	1.00	0.87	1.15	-308	+35	-7,003
60m	25	1.00	1.00	1.00	-169	+52	-1,355

CatBoost (top-5% gate)

h	n_{fire}	$P(\text{fill} \mid \text{move})$			ΔPnL		PnL
		adv	fav	ratio	adv	fav	
40s	545	0.96	0.66	1.46	-97	+52	-9,616
60s	393	0.98	0.70	1.40	-100	+54	-8,119
300s	102	0.98	0.71	1.38	-103	+50	-4,141
600s	55	0.96	0.96	1.01	-102	+50	-1,317
900s	38	1.00	1.00	1.00	+48	+54	+1,966

Why the SDE approach fails: the R_2 drift is noise



ES & VX co-movement



state $\bar{R}_{1,t}, \bar{R}_{2,t}$



R_1 -drift vs. R_2 -drift

PnL by channel, W1, 03-2025 – 05-2025

h	n_{R_1}	Σ_{R_1}	μ_{R_1}	n_{R_2}	Σ_{R_2}	μ_{R_2}
60s	331	+2,572	+8	1,319	-83,002	-63
300s	45	+1,951	+43	339	-19,046	-56
600s	17	-37	-2	195	-11,329	-58
900s	13	-779	-60	140	-10,058	-72
30m	5	+289	+58	76	-9,017	-119

- The R_2 drift initiates most fills and books a negative mean per trade at every horizon. Acts as noise.
- R_1 -drift (mean-reversion) is the only channel that pays, and only in the high-vol Window 1; in calm Window 2 it doesn't overcome R_2 -drift.
- Leverage lives in the diffusion $\beta_1 \bar{\lambda}_1 \sigma_t dW_t$ is a martingale increment that vanishes under $\mathbb{E}[\cdot | \mathcal{F}_t]$. The SDE strategy is blind to it.

Summary

SDE approach

- No directional skill; MAE no better than no-change; forecasts sub-tick.
- Loses PnL almost every cell;
- Why: leverage vanishes under conditioning; the R_2 drift is 1-min noise ; R_1 drift pays only in high vol.

CatBoost approach

- Real, leakage-free directional edge at sub-minute (0.589@40s, 0.577@60s vs. perm. 0.51).
- L_2 loss is suboptimal and subsread. Execution is adversely selected.
- The edge is an order-book effect, not a PDV one.

A sign can be forecast at the sub-minute scale, but not monetized by crossing the spread at one tick rule. The path forward in PnL is market-making frame (two-sided quotes, inventory & adverse-selection control).

Additional result: Theory add-on

The TSPL model admits a Markov form through an infinite continuum of OU factors.

0. Laplace transform

$$(\tau + \delta)^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda^{\alpha-1} e^{-\lambda\delta} e^{-\lambda\tau} d\lambda.$$

1. Kernel representation

$$K_{\alpha,\delta}(\tau) = \int_0^\infty \lambda e^{-\lambda\tau} g(\lambda) d\lambda, \quad g(\lambda) = \frac{\delta^{\alpha-1}}{\Gamma(\alpha-1)} \lambda^{\alpha-2} e^{-\lambda\delta}.$$

2. Markovian system in $\lambda \in (0, \infty)$

$$R_{n,t} = \bar{\lambda}_n \int_0^\infty \frac{F_{n,t}^\lambda}{\lambda} w_n(\lambda) d\lambda, \quad F_{n,t}^\lambda := \int_{-\infty}^t \lambda e^{-\lambda(t-u)} \left(\frac{dS_u}{S_u} \right)^n,$$

$$\bar{R}_{n,t} = \int_0^\infty F_{n,t}^\lambda w_n(\lambda) d\lambda, \quad w_n(\lambda) = \frac{\delta_n^{\alpha_n}}{\Gamma(\alpha_n)} \lambda^{\alpha_n-1} e^{-\lambda\delta_n},$$

$$dF_{1,t}^\lambda = \lambda(\sigma_t dW_t - F_{1,t}^\lambda dt), \quad dF_{2,t}^\lambda = \lambda(\sigma_t^2 - F_{2,t}^\lambda) dt.$$

3. Recover Guyon's $d\sigma_t$ form with $\bar{\lambda}_n = \frac{\alpha_n-1}{\delta_n}$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t.$$

Why it matters

- The n -EXP truncation is in dimension, not structure: 4 factors approximate an infinite-dimensional system.
- Explicit vol-of-vol $\beta_1 \bar{\lambda}_1$; δ_1 is the attention horizon - large δ_1 smooths attention over past returns and damps vol-of-vol. Similar logic applies to memory speed α_1 .

Markovian in

$$(S_t, \{F_{1,t}^\lambda, F_{2,t}^\lambda\}_{\lambda \in (0, \infty)}).$$

Results of this work

- a. Explored the applicability of Guyon's PDV model to VX-futures forecasting and trading. Showed the parametric drift carries no tradable edge.
- b. Analysed the application of machine learning to VX futures. Real, leakage-free sub-minute sign edge that does not yet monetize due to adverse selection.
- c. Enhanced the theoretical model through error analysis and verification fixes. A prior-VIX anchor and intraday/overnight dynamics that aim to cure R_2 -saturation and capture intraday info.
- d. Enhanced the theoretical model by deriving an infinite-OU representation and an explicit vol-of-vol interpretation.

Thank you

Q & A